Team 4 Adaptive Welding Control

Proposed software solution and supporting PINN evidence

1 The problem we aim to solve

During welding, heat can bend the workpiece and move the seam away from its original position. Small setup differences and changes in the welding process can also create a mismatch. A robot following the original path may then miss the seam.

We propose software that estimates this mismatch and suggests a controlled change to the welding path. The same software would suggest a limited travel-speed change when the electrical process readings differ from the planned values.

What the software would produce

- A path-offset proposal tied to the original plan and its revision.
- A hold, pause or replan request when feedback is missing or the required correction is too large.
- An optional travel-speed proposal, with a reason and clear limits.
- A record of the inputs, decisions and tracking errors for review.

How the proposed solution would work

1. Receive the planned seam, process settings and synchronized observations.

2. Check that the readings are valid, recent and expressed in the same coordinate frame.

3. Estimate how far the observed seam has moved from the plan.

4. Smooth the estimate and limit the size and speed of the proposed change.

5. Send the proposal for motion validation and execution approval.

6. Read new feedback and repeat, while recording each decision.

The first version would correct lateral and vertical seam position, called Y and Z. X would remain the planned station because the available research observations do not measure it independently. Torch orientation would stay unchanged until an agreed orientation-update method is available.

Integration boundary

External role	Information exchanged
Path planner	Supplies the original path and revision; checks corrected motion for reachability, joint limits and collisions.
Vision and process monitoring	Supplies seam position, current, voltage, travel speed, signal validity and measurement times.
Execution coordinator	Supplies cell status, approves execution and acknowledges whether a proposal was applied.

The module would produce proposals. Physical motion, robot inverse kinematics, complete scene reconstruction and camera acquisition require their respective external systems.

2 Path and process correction

Estimate the seam deviation

We would compare positions at the same time and nominal station:

Deviation = observed seam position - original planned seam position

A positive Y deviation means that the seam is displaced in the positive Y direction. The proposed target would move in the same direction. Coordinates must share a verified frame; camera coordinates cannot be treated as robot coordinates without calibration.

Smooth the estimate

Sensor noise can make successive readings jump. A low-pass filter would reduce these jumps by combining the latest deviation with the previous estimate:

Filtered offset = (1 - alpha) x previous offset + alpha x measured deviation

The weight would follow **alpha = 1 - exp(-delta_t / tau)**, where delta_t is the update interval and tau controls smoothing. A larger tau gives a slower, smoother response. At one-second intervals, alpha 0.35 gives 65% weight to the previous estimate and 35% to the new deviation. This example explains the filter; operating settings would need separate validation.

Keep the correction within limits

Proposed starting setting	Purpose
Deadband of 0.02 mm	Treat very small deviations as zero to reduce unnecessary movement.
Maximum offset of 1 mm	Request replanning when the measured displacement is beyond the represented correction range.
Maximum offset change of 0.25 mm/s	Prevent a sudden jump between successive proposals.
Timeout of 0.3 s in the 10 Hz test profile	Request a pause when the feedback is no longer recent enough.

These values are development assumptions requiring agreement and testing. They are not approved industrial operating limits.

Proposed seam target = original planned seam target + bounded absolute offset

Each offset would refer to the unchanged original plan. Repeated delivery of the same message must not add the correction again.

Suggest a travel-speed adjustment

We propose a simple electrical-energy-per-length check:

Electrical energy per length = current x voltage / travel speed

With speed in m/s, the quantity is in J/m. If it differs from the nominal input-card value by more than 10%, propose **speed = observed current x observed voltage / nominal energy per length**. Limit the proposal to 4.5-5.5 mm/s and its change to 0.25 mm/s per second. Flag requests outside this range and keep current and voltage setpoints unchanged.

This is an electrical-energy proxy. Its effect on temperature, penetration and weld quality would require a validated process model and physical evidence.

3 PINN assisted prediction

Why include a predictive model

Feedback describes what has already been observed. A Physics-Informed Neural Network, or PINN, could also estimate how heating will move the seam. We propose using this prediction as an initial path estimate, then using fresh observations to check and adjust the remaining mismatch.

The current research model takes workpiece coordinates, time, geometry, material/source settings and a prescribed process history. It predicts temperature and displacement, from which seam and torch target positions are reconstructed. Mapping these inputs to available sensor information remains part of the integration work.

How the model would support the solution

- Use CalculiX to provide thermal and deformation reference data for selected welding conditions.
- Train the temperature network with reference data and a heat-equation penalty, using PyTorch automatic differentiation [1, 2].
- Train the displacement network against the reference deformation and seam targets.
- Keep complete welding runs separate during training and evaluation.
- Freeze the model during evaluation and pass every proposed target through the same downstream motion checks.

The physics part guides the temperature prediction. The displacement component learns from reference deformation data; it does not solve the full workpiece mechanics.

Available PINN evidence

The saved evaluation contains seven held-out synthetic welds and 69 active one-second seam samples. The reference is the displaced seam calculated by CalculiX.

Seam tracking comparison	RMSE in mm
Original path without correction	0.047196
Predicted seam target from the PINN	0.019943

Reduction in seam RMSE: approximately 57.7%.

RMSE is the square root of the mean squared distance between a proposed target and its reference. Smaller values mean closer tracking. The reduction is calculated as 100 x (1 - PINN RMSE / original-path RMSE).

These results support further work on prediction-assisted correction within the represented synthetic conditions. They do not measure a physical robot or prove that the proposed feedback controller will achieve the same reduction. The model uses exact simulation inputs, and its advantage over other neural approaches has not been established.

Alternatives to compare

Approach	Proposed role
Filtered measurement correction	A simple baseline with visible rules and limits.
PINN assisted correction	Predict thermal displacement and use observations to check the prediction.
Model Predictive Control	A later option for choosing several future corrections together, after the plant model and constraints are validated.

4 Proposed inputs and outputs

The interface would use JSON so that the module can be tested independently and connected to other software. JSON Schema would check message structure and field types [3]. The formats remain proposals until producers and consumers agree on their meaning.

Shared conventions

- Positions and offsets in metres, time in seconds and angles in radians.
- An explicit coordinate frame, initially the workpiece frame.
- A trajectory ID and revision identifying the unchanged original plan.
- Measurement and receipt times on a common clock; generation metadata would use UTC.
- Unavailable measurements, confidence and quality information represented by null.
- A validity decision, reason and expiry that the consumer checks before using an offset.

Message purpose

Message	Proposed contents
target_path.json	Planned and predicted geometric targets, units, frame and prediction source.
correction.json	Absolute offset, optional process proposal, action, validity, reason and plan revision.
correction_history.json	Ordered proposals and rejection events for replay.
metrics.json	Tracking measures where a reference exists, input availability and processing time.

Illustrative message fields

The following extracts show the idea using invented values. They are not complete schema-valid messages or measured results; full messages would include identifiers, source information and validity details.

Input extract:

```
{
  "trajectory_revision": "plan_v1",
  "clock": {"kind": "simulation", "now_s": 1.0},
  "sample": {
    "frame": "workpiece",
    "planned_seam_m": [0.020, 0.000, 0.003],
    "observation": {"t_s": 1.0, "visible": true,
      "measured_axes": ["y", "z"],
      "seam_yz_m": [0.00010, 0.00300]}
  }
}
```

Output extract for a first filtered update at the illustrated one-second cadence:

```
{
  "trajectory_revision": "plan_v1",
  "status": "VALID", "action": "OFFSET_PATH",
  "correction": {"dx": 0.0, "dy": 0.000035, "dz": 0.0},
  "confidence": null,
  "actual_execution_enabled": false
}
```

The positive Y offset follows the observed positive Y displacement. A retained offset in a rejected or expired message would not authorize motion. Execution acknowledgement would be recorded separately from the proposal.

5 Timing and validation approach

What update frequency means

Frequency is the number of updates per second. At 10 Hz, a new calculation is requested every 0.1 s. At 100 Hz, the interval is 0.01 s. This is separate from the robot's travel speed in mm/s.

Signal or activity	Proposed treatment
Existing research path data at 1 Hz	Use for offline model evaluation; do not treat interpolated points as new physical observations.
Initial controller target at 10 Hz	Test the complete read-check-calculate-propose cycle every 0.1 s.
Possible path/process updates at 10-100 Hz	Consider higher rates only after end-to-end timing and input freshness are demonstrated.
Vision feedback around 5-30 Hz	Confirm the actual camera output rate and delay before selecting operating settings.

A faster correction loop does not create new camera information. Each reading needs its own measurement time and age check. We would measure controller calculation time separately from camera capture, transport, planning checks and execution acknowledgement.

Planned evaluation

We would define the test conditions before comparing approaches and tune settings on development data. Complete held-out welding runs would remain untouched during tuning.

Test condition	Behaviour to check
No deviation	Keep the nominal path and process settings.
Step or gradual seam movement	Follow the change while respecting offset and rate limits.
Noisy observations	Compare tracking error, jitter and response delay.
Missing or delayed feedback	Hold only within the freshness allowance, then request a pause and check recovery.
Large deviation or invalid message	Reject an unsuitable correction and request replanning where required.
Duplicate message or changed plan	Avoid repeated accumulation; require the correct plan revision and state.

Measures to record

- Seam and torch-target RMSE, maximum error and percentage within an agreed tolerance.
- Correction magnitude, rate, jitter and settling time after a disturbance.
- Availability of usable feedback and counts of holds, rejections and replan requests.
- Calculation latency, complete cycle delay and missed update deadlines.
- Valid JSON output, correct units, correction sign and consistent replay behaviour.

Where no independent reference is available, tracking accuracy would remain unavailable. Weld quality would require measured indicators such as bead geometry or inspection labels; path error alone would not supply a quality score.

6 Tools and implementation plan

Proposed software tools

Tool	Why we would use it
Python and NumPy	Implement correction rules, handle coordinate arrays and prepare repeatable tests.
PyTorch	Train and run the neural model and calculate derivatives for the physics loss.
CalculiX	Produce thermal and deformation references for the research dataset.
JSON Schema and a local API	Check data formats and exchange proposals with other software.

ROS 2 could be added as a communication adapter if the shared integration environment requires it. The correction rules and interface meaning would remain independent of the transport choice.

Suggested work sequence

Stage	Work and acceptance evidence
Agree on interfaces	Confirm units, frame transforms, timestamps, revisions and proposal acknowledgements with the input/output owners.
Establish the baseline	Implement the filtered correction and process rule, with explicit limits and failure responses.
Compare prediction methods	Evaluate measurement-only and PINN assisted approaches using comparable available information and fixed held-out data.
Integrate software	Demonstrate that a consumer accepts or rejects each proposal and records the outcome.
Validate physical operation	Confirm approved limits, sensing accuracy, timing, motion feasibility and process response before robot use.

Current blockers

Image dataset: The required welding image dataset has not yet been received. Representative images and agreed seam labels are needed to validate the use of image-based feedback.

Integration inputs: We need actual feedback signals and their accuracy, a calibrated coordinate transform, the robot/TCP definition, approved correction and process limits, and an acknowledgement and recovery protocol. Quality-related adjustment also needs reliable quality measurements and an agreed response policy.

The initial scope is a translation proposal with limited speed adaptation. Full pose correction, online learning, predictive optimization and physical quality control would be added only when the required inputs and validation evidence are available.

References

[1] CalculiX project documentation. https://www.calculix.de/

[2] PyTorch automatic differentiation documentation. https://docs.pytorch.org/docs/stable/autograd.html

[3] JSON Schema overview. https://json-schema.org/overview/what-is-jsonschema

PINN benchmark source: saved held-out path metrics in phase2/output/metrics_comparison.json. These numbers come from the existing evaluation; no new experimental results are claimed.